# Myelin: Secure Enterprise Integration Layer for Neurotechnology

*Venture Proposal, Zell Entrepreneurship Program - 2026 batch application / Omer Shalev*

---

**One-Liner:** *Secure, hardware-agnostic integration layer bringing privacy-first Brain-Computer Interfaces to enterprises and applications.*

### 1. The Problem: Enterprise adoption barriers in non-invasive BCI

Brain–computer interfaces (BCIs) are moving from research into real deployments, led by non-invasive EEG headsets that translate brain activity into software-ready signals (e.g., attention, fatigue, intent). Market forecasts reflect that momentum: the global BCI market is projected to grow from about $2.94B in 2025 to $13.86B by 2035 (~16.77% CAGR) [1], with non-invasive systems already representing most revenue, about 82% (~$2.41B in 2025) [2]. Even so, enterprise-wide rollout remains early, with adoption clustering first in safety and fatigue-critical environments like aviation and trucking, where incremental improvements have outsized operational value (i.e., [3]).

Setting cost aside (which is improving dramatically year over year), **two main barriers** keep non-invasive BCIs from scaling across wider adoption within work environments and enterprises: **1) Hardware fragmentation, making development and deployment expensive**: Today's BCI ecosystem is siloed. Each headset vendor (e.g., Emotiv, OpenBCI, Neurable, NextMind) ships its own interface and data formats, so a team building one "focus" or "fatigue" app ends up maintaining multiple device integrations (across devices with different channel counts, signal quality, and noise characteristics). For enterprises and developers, that means fleets can't mix brands or switch suppliers without custom engineering, creating vendor lock-in, slow rollouts, and high maintenance costs. **2) Privacy and regulation**: Raw EEG is not just "sensor data"; it's neural data that can support sensitive inferences about health, emotion, and cognition, which raises the stakes for consent, storage, security, and secondary use. Regulation is tightening quickly and remains in flux, for instance, in early 2025 California updated the CCPA to explicitly include "consumer's neural data" as sensitive personal information [4]. It raises the barrier to entry for builders and startups, and it makes enterprise deployments harder to approve and scale.

### 2. The Solution & UVP: One Integration Layer, Privacy and security by design

We're building a smart, secure, privacy-preserving integration layer between diverse BCI headsets and the applications that use them, **dramatically lowering the barrier to building new application-layer BCI products**. Raw EEG stays on-device and is processed inside a secure, trusted edge compute environment: we run device-aware preprocessing and denoising, normalize across different sensors and channel counts, and convert the signal into privacy-protected features. We apply Differential Privacy to reduce biometric "signature" leakage, and when external compute is required, we use cryptographic techniques such as homomorphic encryption so analytics can run without exposing the underlying signal. Only anonymized, minimal, actionable outputs (e.g., focus level, stress) leave the device. Developers never touch raw neural data, enterprises never store biometric identifiers, and compliance is enforced by default in every integration (just like an API).

### 3. Market & Business Model

Rather than relying on top-down hardware sales projections, our market sizing targets the massive friction in BCI software R&D and software-hardware integration. Globally, an estimated 3,000+ neuro-startups, academic labs, and enterprise fleets [5], waste over $450M annually on in-house engineering and legal compliance to integrate fragmented BCI hardware[1]. By capturing roughly 10% of these early adopters, we target a highly defensible $45M Serviceable Obtainable Market (SOM), scaling our revenue by monetizing the exact engineering time and compliance costs we eliminate for our customers.

Our business model is B2D (Business-to-Developers), aiming at:

- Enterprises: Enterprises, especially safety-critical fleets (trucking, mining), will use Myelin to unify cognitive signals across mixed-brand headsets, reducing incidents while protecting workforce privacy through on-device trusted processing.
- BCI startups, neuro-integrated apps & gaming studios: Developers will license Myelin as "BDaS" (Brain Data As a Service), "Compliance-in-a-Box," integrating biosignals like stress and focus without building custom hardware drivers, implementing complex neuro-algorithms, or managing raw neural data exposure.

---

[1] SOM Calculation: 3,000 active global BCI groups × 1 dedicated integration engineer @ $150,000/yr = $450M total R&D friction. Before talking about legal aspects and more.

- Research & clinical labs: Universities and hospitals aggregate multi-headset datasets out of the box and streamline IRB/privacy approval by keeping sensitive processing at the edge and exporting only approved outputs.

Unified revenue model (Hybrid SaaS): We charge on a single Monthly Active Connections (MAC) metric: customers pay a fixed platform subscription (SDK + dashboard + compliance controls) plus a per-connection usage fee, scaling cleanly from hundreds of workers (within enterprises) to tens of thousands of consumers in neuro-integrated consumer applications (**very similar to Plaid model in finances**).

### 4. The Moat & Why Me

We are not competing with hardware makers; we are their ultimate distribution channel. The more headsets we support, the more developers build on us. The more developers build on us, the more enterprises adopt our platform. This creates a self-reinforcing network effect: each new integration makes the platform stickier for every participant in the ecosystem.

I bring a rare combination that's directly relevant to this problem: over a decade in technology and security, including cyber & cryptography roles in an elite unit within the Prime Minister's Office, plus hands-on neuroscience research at Prof. Amir Amedi's lab, with EEG and fMRI across multiple acquisition setups and analysis pipelines.

### 5. Roadmap & Go-to-Market

- **Phase 1 — Build**: Raise a small initial round and ship the core privacy-first integration layer: signal normalization, preprocessing, a plug-and-play framework for neuro-algorithm integration, and a Differential Privacy module. Obtain a formal legal opinion covering new legislation (i.e., CCPA), and GDPR, both to validate compliance and to establish a defensible moat.
- **Phase 2 — B2D Launch:** Go-to-market as a B2D (Business-to-Developer) platform targeting neurotech application-layer startups (300+ globally) and university/research institutions operating on grant funding (a price-insensitive buyer with immediate integration pain). Raise a seed round on early traction.
- **Phase 3 — Expansion (Enterprise & Gaming):** Expand into enterprise workforce safety deployments and gaming/XR studio integrations, leveraging the developer ecosystem and headset coverage built in Phase 2 to drive top-down enterprise sales and platform-level partnerships.